

Mission: to help you treasure and share the beautiful experiences these pieces of art gave you, and connect you with more

When we watch a movie, especially a good one, we are reminded of memories, experiences, we are moved, we start thinking. Even the name of a movie we loved can send us back, giving us a wonderful feeling of comfort, joy, and appreciation for the beauty in this world.

To Do:

1. never provide recommendations that are the same as the movies inputted (the whole point of recommendations are movies which the user didn't input)
2. create a loading page or animation while the recommendations are being created. this loading should have some sort of way to show the user how much has been done and how much is left (ie % completed and time left with a bar or something like that)
3. write a new readme in markdown .md format (replacing all the code blocks with [----]) for the new code
4. Create a user system where you can sign up/sign in with a username and password and the system stores your recommendations, and you can later access and add to them
5. Create a system to help the ai learn from its recommendations
 - a. Enable the user to thumbs up or thumbs down each recommendation, feedback which the ai can learn from (either for that user or for the entire system?)
6. If there are no recommendations, say so
7. There should be a way to delete a recommendation
8. What should the recommendation be out of? 5 with .5 increments? why?
9. Fix the lengthy startup time for the system
10. Filter by genre, media type, or maturity rating
11. Show movie/TV posters
12. Save user profiles and rating history
13. Pagination and mobile optimization
14. please make sure in the search bar, the release date of the movie appears with the suggestions because there are several movies with the same names and the user needs a way to distinguish them somehow.

Current summary of my code generated by an ai chatbot (I will say that it used to work a lot better without the TMDB stuff, I am coming back to this after a while of not working on it, i forget the previous code of just the MovieLens data set exactly where that was and i have added changes I do want to keep since then, but Im not quite sure why I have TMDB in this):

Project Overview

You are building a full-featured movie recommendation web app using Python (Flask), HTML/CSS, and supporting libraries. The app:

- Uses both content-based filtering and collaborative filtering for hybrid recommendations.

- Allows users to create accounts, maintain personal movie libraries, and generate personalized recommendations.
 - Supports a temporary one-off recommendation feature for non-saved exploration.
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✓ Features You've Implemented

1. 🗝️ User Authentication System

- Login & Signup pages created (login.html).
 - Secure user data stored in a local SQLite database (users.db).
 - Passwords are hashed using werkzeug.security.
 - After login/signup, users are redirected to the Menu Page (/menu), not directly to recommendations.
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2. 🏠 Menu Page ("Home")

- After logging in, users see a centralized Menu Page (menu.html) with:
 - App title (currently a placeholder, e.g. "Insert App Name Here").
 - Three buttons:
 - Edit Library
 - Generate Recommendations Using Library
 - One-Off Recommendations (name undecided — suggestions pending)
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3. 📖 Library Page (/library)

- Lets the user build a custom movie library:
 - Add one movie at a time with a rating (autocomplete + rating selection).
 - Movies are stored per user in the database.
 - Ratings can be edited and movies can be deleted.
 - Library updates are only saved after clicking a "Save Library" button.
 - This button is grayed out unless changes are made.
 - If a user tries to leave without saving, they're prompted to confirm or save.
 - The library is sorted most-recent-first, and re-adding a deleted movie puts it back on top.
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4. 🎯 Recommendations from Library (/recommend_from_library)

- Generates personalized hybrid recommendations from the saved library.
- Uses collaborative filtering (MovieLens ratings) + content-based features (genres, etc.).

- Displays both the recommendations and the user's saved library below.
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5. 🌐 One-Off Recommendations Page (/one_off_recommend)

- Allows a user to temporarily input movie ratings.
 - These are used only for this session.
 - Ratings are not saved, and do not affect the library.
 - Useful for generating recommendations for a small number of one-time inputs.
 - You're still deciding on a better name for this feature. (Suggestions include: "Quick Pick," "Just This Time," "Ad Hoc Recs," "Explore Mode.")
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6. 📦 Data Handling

- Movies are loaded from a TMDb-enhanced metadata file.
 - Collaborative filtering uses the MovieLens ratings.csv dataset (preprocessed).
 - Title search is case-insensitive, shows year in autocomplete, and excludes non-movie/TV entries.
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7. 🖥️ Front-End Behavior

- Pages use basic HTML templates (index.html, login.html, menu.html, etc.) stored in a templates/ folder.
 - Includes a loading animation with progress bar when generating recommendations.
 - Handles user feedback (e.g. warnings on unsaved changes).
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⚠️ Past Bugs/Resolved Issues

- 🟢 Duplicate years in titles — fixed by relying on TMDb-provided titles that already include the year.
 - 🟢 Mismatch between collaborative/content scores — fixed by filtering recommendation pool.
 - 🟢 Recommender not loading — caused by JavaScript not triggering a POST request — fixed.
 - 🟢 Redirecting to /recommend instead of /menu after login — fixed by changing redirect logic.
 - 🟢 Duplicate movies in saved library — now prevented at the database logic level.
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Planned/Optional Next Steps

You've expressed interest in adding:

-  This was the next task: Temporary recommendation mode page (now done).
-  Refine and name the app + "One-Off" feature.
-  Add filters (genre, type, maturity rating) to sort results.
-  Allow 1–100 scale ratings instead of 1–5.
-  Let users view/export past recommendations.
-  Style enhancements for layout polish and responsiveness.

Past messages to an ai chatbot that was helping me code:

Uggg, this is really complex. basically, I want to go back to the point where I was just using movies from movies.csv and ratings.csv. I liked the previous UI (<https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css>). i want the application to fulfill these points:

hmmm, basic functionality wise that seems to be working pretty well. however, i see one problem and one thing i want to do next. the problem is that right now, when i enter a movie in, it stores it in the past recommendations correctly. but, when i enter it in again, it adds that movie again to past recommendations (duplicate).

the thing i want to do next is that right now i have one recommender page and the system just stores the movies in past recommendations and does nothing with it. i want to have one page where the user can add movies with ratings one at a time which are stored in something called "the library".

after the user logs in, the program should display a menu screen. right above the center of the screen should be large words saying "insert app name here" (i haven't decided on a name for this program yet, but want a placeholder). below that (around the center of the page and on the left half under the app name placeholder should be a "edit library" button. below that should be a "generate recommendations using library" button. on the right half under the app name placeholder should be a button. when the user clicks edit library, the program should bring the user to the library page which contains all of past movies the user has added and their ratings (and say "No movies added yet, add some to generate recommendations!" if there are none added yet). above all of the movies added and their ratings, there should be a place to add movies to the library (you should only be able to add one movie at a time tho, meaning after a

movie is found and rating given, the user should have to click an "add movie" button which brings it down to the top of the library which is displayed right beneath it). the user should also be able to edit ratings and remove movies added from the library. there should be a button at the top called "save library" which updates the movies added to the current version of the library and brings the user back to the menu page. the "save library" button should be grayed out until changes are made, and if the user tries to/accidentally tries to leave the page without saving, the program should display a message saying changes have not been saved, with an option to save. if the user clicks the "generate recommendations using library" button, the system should generate recommendations based on the movies and ratings in the most recently saved version of the library. underneath these recommendations should be the library (so the user can remember what they have added). the library should always be ordered so that the most recently added movie is at the top and the oldest added movie is at the bottom. if a movie's rating is changed, their place in the library order shouldn't change after the user saves the changes but if, for example, the movie is deleted, then the library saved, then the user adds that same movie back later, it should appear at the top again as if it had never been added. that is a fringe case tho.

the unspecified button from before in the center of the menu page on the right half under the app name placeholder should bring the user to a recommendations page. this page should allow the user to add movies and ratings and produce recommendations from those movies added. however, none of these movies should be stored in the library. this button is so that if, for example, the user has added a good amount of movies to the library, but has several specific movies that they want recommendations for, they can do so without having to delete all of their past ratings in the library. this page only looks at the movies and ratings added now, does not save any ratings added after the user leaves the page, and does not use any past ratings added when generating recommendations unlike the generate recommendations from library page. i still don't have a good name for this page and button, but if you could also provide a couple ideas for it, that would be great.